

- 7 (a)** State experimental evidence to suggest that the process of radioactive decay is
- (i)** random;

[1]

(ii) spontaneous.

[1]

- (b) A student measured the count rate from a sample of pure iodine-131 at various times. The results are shown in Fig. 7.1. Iodine-131 decays by beta emission to form xenon-131 which is stable.

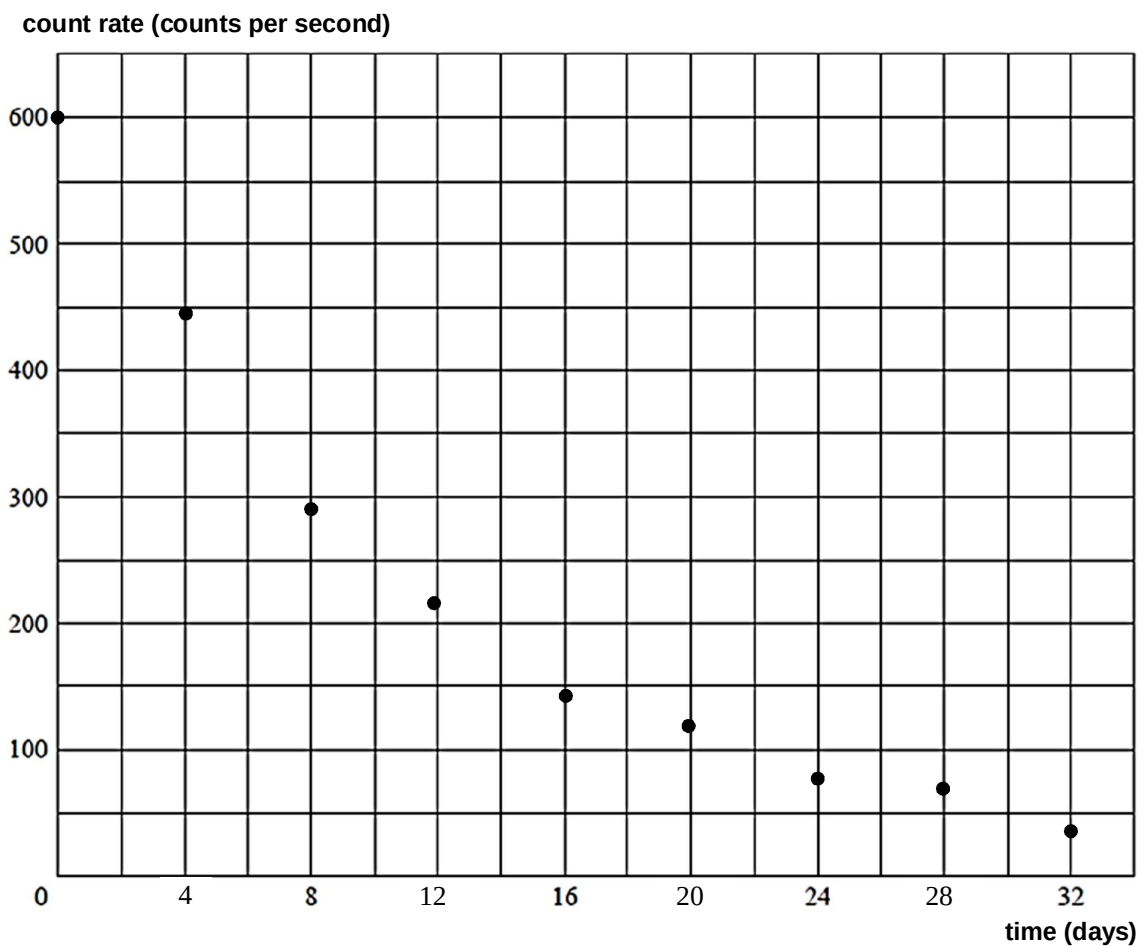


Fig. 7.1

Use Fig. 7.1 to determine a value for the half-life, in days, of iodine-131.

half life = days [3]

(c) Explain why the determination of the half-life of iodine-131 by the method in **(b)** requires that the product of the decay is stable.

[1]

(d) Suggest why the percentage systematic error will not be significant if the student neglected the background radiation.

[1]